When Are Sufficient Statistics Sufficient? Bounding Gains-from-Trade Formulas with an Application to NAFTA *

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Abstract

The welfare formula of Arkolakis et al. (2012) (ACR) is applied often through two classes of models, Eaton–Kortum (EK) and Melitz, both of which are built on specific distributional assumptions: Fréchet in EK, Pareto in Melitz. When these assumptions are violated, the four ACR macro-restrictions still hold, but the trade elasticity is no longer constant, and the ACR formula, which assumes a single elasticity, acquires an approximation error. We derive strict upper bounds on this error for both classes. The bounds depend only on the curvature of the gravity equation and are estimable without firm-level data. Applied to the NAFTA calculation of Caliendo and Parro (2015), the Fréchet assumption introduces non-negligible uncertainty concentrated in agriculture but does not overturn their conclusions.

JEL Codes: F10, F14, C52

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*Replication package is available at https://github.com/yonggeun-jung/ACR_2012_approximation_public. The package includes all code and processed master files sufficient to reproduce all tables and figures. The Caliendo and Parro (2015) replication package is available from https://academic.oup.com/restud/article/82/1/1/1547758; raw data (CEPII Gravity database and BACI) from http://www.cepii.fr. First version: March, 2026.

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1 Introduction

Arkolakis et al. (2012) (hereafter ACR) establish one of the most influential results in modern trade theory: a broad class of models deliver the welfare formula:

$$\hat{W} = \hat{\lambda}^{-\frac{1}{\varepsilon}},$$

where $\hat{W}$ is the change in real income, $\hat{\lambda}$ is the change in the domestic expenditure share, and ε is the trade elasticity. The elegance and generality of this formula have made it a centerpiece of quantitative trade analysis. It serves as the primary welfare benchmark in evaluations of trade liberalisation (e.g., Costinot and Rodríguez-Clare, 2014), counterfactual analyses of regional trade agreements (Caliendo and Parro, 2015), and assessments of trade policy at the sectoral level (Ossa, 2015). Its appeal is that welfare calculations reduce to two observable objects, $\hat{\lambda}$ and ε, without requiring the full structure of the underlying model.

The two models most commonly used to apply the ACR formula, Eaton–Kortum (EK, Eaton and Kortum, 2002) and Melitz (Melitz, 2003), are themselves built on distributional assumptions: Fréchet productivity in EK, Pareto productivity in Melitz. When these distributional assumptions are violated, the four ACR macro-restrictions[1] still hold, but the trade elasticity varies along the equilibrium path. Because the ACR formula evaluates welfare changes using a single, constant elasticity, this variation translates directly into an approximation error. The empirical relevance of this concern is well established: evidence that productivity distributions depart from Fréchet and Pareto is extensive (e.g., Di Giovanni and Levchenko, 2013; Head et al., 2014; Bas et al., 2017; Fernandes et al., 2023), and Melitz and Redding (2015) show that non-Pareto distributions generate new welfare margins absent from the ACR formula. Researchers who rely on the formula are implicitly assuming that the distributional departure is small enough to be innocuous, but the existing literature provides no way to evaluate that assumption: given a particular dataset, how large could the ACR approximation error actually be?

This paper provides an upper bound on the ACR approximation error that researchers can compute alongside any welfare calculation. We hold the four ACR macro-restrictions fixed and ask how large the error becomes when only the distributional assumption is relaxed. We derive strict upper bounds for both the EK and Melitz model classes. The bounds depend on a single macro-observable quantity: the variation of the distance

[1] For the four restrictions, see Section 2. We maintain them throughout and relax only the distributional assumption.

elasticity of trade across bilateral pairs. We validate the bounds through Monte Carlo simulations across four data-generating processes. As an application, we evaluate how much the distributional assumption matters for the influential NAFTA welfare calculation of Caliendo and Parro (2015) (hereafter CP), whose multi-sector EK model uses the sectoral ACR formula as a building block. The aggregate error bound for Mexico is 0.10 percentage points, compared with CP's reported welfare gain of 1.31%, implying that the Fréchet assumption does not overturn the qualitative conclusion but contributes non-negligible uncertainty concentrated in agriculture and primary products.

Our contributions are as follows. First, we make explicit the logical position of the distributional assumption within EK and Melitz: the four ACR macro-restrictions hold in these classes only in conjunction with Fréchet or Pareto productivity, so the distributional assumption operates as a fifth requirement when the abstract ACR class is instantiated through its two leading quantitative implementations. This reframes the Fréchet and Pareto assumptions, often treated as modeling conveniences, as load-bearing structural restrictions responsible for the approximation error. Second, by recasting the micro-level bounds into macro-observable equivalents, we make this fifth requirement testable without firm-level data and without abandoning the closed-form structure of the ACR formula. Third, the local sufficiency conditions we derive (Corollaries 1 and 2) clarify when the distributional assumption is innocuous and when it is not, providing quantitative content to complement the qualitative warnings in Head et al. (2014) and Melitz and Redding (2015).

Two strands of the existing literature relate closely to ours. Nigai (2017) shows using firm-level data that neither Pareto nor log-normal alone captures the productivity distribution, and computes welfare gains under a specific Pareto–log-normal mixture; we provide instead a distribution-free upper bound that applies regardless of the functional form of the departure. Adao et al. (2017) sidestep the distributional assumption altogether, developing a nonparametric counterfactual methodology that recovers welfare predictions from a reduced factor demand system; this approach generalizes the parametric gravity framework but requires estimating the demand system. We take a different route: we retain the ACR closed-form formula, preserving the tractability that motivated its widespread adoption, and instead bound the error introduced when its distributional assumptions are violated.

Our scope is also deliberately narrower in another direction. Violations of the four ACR macro-restrictions themselves, most notably variable-markup environments (e.g., Atkeson and Burstein, 2008; Arkolakis et al., 2019; Firooz and Heins, 2026) that fail the constant-profit-share condition, require alternative sufficient statistics beyond $(\hat{\lambda}, \varepsilon)$ and

welfare formulas that are not derived from the ACR representation. We leave such extensions to companion work and focus here on distributional misspecification within the EK and Melitz classes.

The remainder of this paper is organized as follows. Section 2 introduces the framework and defines the approximation error. Section 3 presents the theoretical bounds and their macro-observable reformulations. Section 4 describes the gravity-based diagnostic. Section 5 validates the bounds through Monte Carlo simulations. Section 6 applies the diagnostic to CP's NAFTA welfare calculation. Section 7 concludes. Appendices report proofs and additional results.

2 Framework

We consider EK-class and Melitz-class trade models satisfying the four ACR macro-restrictions:

(i) CES preferences with elasticity of substitution $\sigma > 1$,

(ii) linear cost structure with iceberg trade costs $\tau_{ij} \geq 1$,

(iii) balanced trade, and

(iv) constant profit share.

Let $\lambda \in (0, 1)$ denote the domestic expenditure share. Under conditions (i)–(iv), the exact welfare change satisfies the integral representation:

$$\ln \hat{W} = -\int_{\ln \lambda}^{\ln \lambda'} \frac{1}{\varepsilon(x)}\, dx, \tag{1}$$

where $\hat{W} \equiv W'/W$ is the ratio of post- to pre-shock real income and $\varepsilon(x) > 0$ is the trade elasticity at domestic share e^x.[2] Importantly, conditions (i)–(iv) do not by themselves pin $\varepsilon(x)$ to a constant. Constancy of ε is an independent property that, in the EK and Melitz classes, is delivered by the distributional assumption defining those classes: *Fréchet* in EK (shape θ) and *Pareto* in Melitz (shape κ). We refer to this requirement as the *distributional assumption* throughout. It is constitutive of the two classes: in EK the Fréchet draw delivers the gravity equation, and in Melitz the Pareto assumption closes the model in

[2]Equation (1) follows from integrating the differential representation in Lemma 1 of Arkolakis et al. (2012) along the equilibrium path; see also Costinot and Rodríguez-Clare (2014).

closed form. When the distributional assumption holds, $\varepsilon(x) = \varepsilon_{\text{base}}$ for all x ($\varepsilon_{\text{base}} = \theta$ in EK, $\varepsilon_{\text{base}} = \kappa$ in Melitz), and Equation (1) collapses to the closed-form ACR formula:

$$\ln \hat{W}_{\text{ACR}} = -\left(\frac{1}{\varepsilon_{\text{base}}}\right) \ln \hat{\lambda},$$

where $\hat{\lambda} \equiv \lambda'/\lambda$. The approximation error studied in this paper is the gap between the exact representation and the ACR formula:

$$\mathcal{E} \equiv \ln \hat{W}_{\text{exact}} - \ln \hat{W}_{\text{ACR}} = \int_{\ln \lambda}^{\ln \lambda'} \frac{\varepsilon(x) - \varepsilon_{\text{base}}}{\varepsilon_{\text{base}} \cdot \varepsilon(x)} \, dx, \tag{2}$$

measured in log points of real income. By construction, $\mathcal{E} = 0$ when the distributional assumption holds. The bounds derived in Section 3 characterize $\mathcal{E}$ for arbitrary distributions consistent with the four macro-restrictions.

We use the following terminology throughout. The *ACR macro-restrictions* are conditions (i)–(iv); the *distributional assumption* is the Fréchet (EK) or Pareto (Melitz) requirement that, together with the macro-restrictions, makes the trade elasticity constant; and the *ACR formula* is the closed-form expression $\hat{W} = \hat{\lambda}^{-1/\varepsilon}$, which holds exactly only when both are satisfied. Our analysis holds the macro-restrictions fixed and quantifies the error introduced when only the distributional assumption is violated.

2.1 EK Class

The trade elasticity depends on the technology distribution $F_0(z)$ with density $f_0(z)$, where $z > 0$ is a firm's efficiency draw. Under Fréchet ($F_0(z) = \exp(-z^{-\theta})$, shape $\theta > 0$), $\varepsilon = \theta$ is constant. Under a general F_0, the elasticity at margin z is the scaled hazard rate:

$$\varepsilon(z) = \frac{z \, f_0(z)}{1 - F_0(z)}, \tag{3}$$

and the hazard-rate deviation from Fréchet is $\eta(z) \equiv \varepsilon(z) - \theta$. The derivation connecting Equation (3) to the structure of the EK model is provided in Appendix A.1.

A useful feature of the EK class is that condition (iv) holds automatically regardless of the underlying distribution: with no entry or exit margin, the equilibrium profit share is pinned down by the CES demand structure alone. The bounds derived below therefore apply to arbitrary F_0 satisfying Assumption 1, which is the relevant case for the multi-sector EK application in Section 6.

2.2 Melitz Class

Firms draw productivity $\varphi > 0$ from $G(\varphi)$ with density $g(\varphi)$. The trade elasticity at export cutoff φ^* is:

$$\varepsilon(\varphi^*) = (\sigma - 1) + \frac{\varphi^* g(\varphi^*)}{\Phi(\varphi^*)}, \tag{4}$$

where $\Phi(\varphi^*) \equiv \int_{\varphi^*}^{\infty} (\varphi/\varphi^*)^{\sigma-1} g(\varphi)\, d\varphi$ is the revenue-weighted productivity aggregate. Under Pareto ($\bar{G}(\varphi) = (\varphi_{\min}/\varphi)^{\kappa}$, $\kappa > \sigma - 1$), $\varepsilon = \kappa$ is constant; this closed-form structure is the basis of the Melitz–Pareto gravity representation in Chaney (2008). Under a general G, the elasticity varies with φ^*. The derivation is provided in Appendix A.2.

Melitz and Redding (2015) note that non-Pareto distributions can introduce welfare margins through the endogenous mass of entrants M_e. We treat the four ACR macro-restrictions as maintained assumptions: under condition (iv), the equilibrium adjustment of M_e is absorbed into $\Phi(\varphi^*)$, so within this restricted class the variation in $\varepsilon(\varphi^*)$ alone captures the approximation error. Distributions for which the entry margin generates first-order departures from constant profit share, such as those producing variable markup dispersion, lie outside the scope of our bound.

The class of non-Pareto distributions consistent with constant profit share is non-empty. The truncated Pareto distribution with upper bound $\bar{\varphi}$, used as DGP (iv) in Section 5, is a leading example: as $\bar{\varphi} \to \infty$, the entry margin's contribution to the profit share vanishes at rate $O(\bar{\varphi}^{-\kappa})$ (Lemma 1), so condition (iv) holds to arbitrary precision for sufficiently high truncation. More generally, any distribution whose departure from Pareto is concentrated away from the equilibrium entry margin satisfies condition (iv) approximately.

3 Theoretical Bounds

The approximation error $\mathcal{E}$ defined in Equation (2) depends on how much the trade elasticity ε deviates from its benchmark value along the equilibrium path. In this section, we formalize this dependence. We first derive bounds in terms of distributional primitives (Section 3.1), then reformulate them in terms of the observable trade elasticity alone (Section 3.2), and finally characterize the rate at which distributions converge to sufficiency (Section 3.3).

3.1 Micro-Level Bounds

EK bound. In the EK class, the trade elasticity $\varepsilon(z) = \theta + \eta(z)$ varies with the marginal technology draw z. The question is: how much does this variation inflate the welfare error? Let $[z_L, z_H]$ denote the range of marginal draws traversed along the equilibrium path induced by the policy shock. We require a mild regularity condition:

Assumption 1. *$\varepsilon(z) \geq \sigma - 1$ for all $z \in [z_L, z_H]$, and $\theta \geq \max\{\sigma - 1, 1\}$.*

Under Fréchet, the price index converges if and only if $\theta > \sigma - 1$ (Eaton and Kortum, 2002). EK extend this regularity condition to general distributions: it requires the trade elasticity to remain at least $\sigma - 1$ along the equilibrium path. This is economically natural. If $\varepsilon(z) < \sigma - 1$ at some margin, trade at that margin is less responsive to trade costs than the intensive margin alone ($\sigma - 1$), which is inconsistent with the comparative advantage mechanism that drives trade in the EK model. The auxiliary lower bound $\theta \geq 1$ is required for the proof of Proposition 1 (Appendix A.3); it is automatically implied by $\theta \geq \sigma - 1$ whenever $\sigma \geq 2$, which covers all standard quantitative applications. The condition is satisfied by all distributions in the domain of attraction of Fréchet with $\theta > \sigma - 1$, including all four DGPs considered in Section 5.

Proposition 1 (EK Bound)**.** *Under the four ACR conditions, Assumption 1, and Fréchet (shape θ) as benchmark:*

$$|\mathcal{E}_{\text{EK}}| \leq \sup_{z \in [z_L, z_H]} \frac{|\eta(z)|}{\theta + \eta(z)} \cdot |\ln \hat{\lambda}|. \tag{5}$$

See Appendix A.3 for the proof, which applies the triangle inequality to Equation (2) directly.

The bound has a transparent structure. The first factor, $\sup |\eta|/(\theta + \eta)$, measures the worst-case proportional hazard-rate deviation along the equilibrium path. It equals zero under Fréchet and increases with the magnitude of the departure. The second factor, $|\ln \hat{\lambda}|$, scales the error by the size of the policy shock: larger shocks traverse a wider segment of the path, exposing the calculation to a greater range of elasticity variation.

Corollary 1 (EK Local Sufficiency)**.** *If $\eta(z) = 0$ for all $z \in [z_L, z_H]$, then $\mathcal{E}_{\text{EK}} = 0$ regardless of the global shape of F_0.*

Corollary 1 states that the ACR formula can be locally exact even if F_0 departs from Fréchet globally. What matters is whether the departure occurs in the region affected by the policy shock. A distribution that is non-Fréchet in the left tail but Fréchet-like in the

right tail will generate zero ACR error for shocks that only move cutoffs within the right tail.

Melitz bound. The Melitz class introduces a richer structure because the trade elasticity in Equation (4) depends on the ratio of two objects: the local density $g(\varphi^*)$ and the revenue-weighted tail aggregate $\Phi(\varphi^*)$. To isolate the sources of departure, we decompose each relative to its Pareto counterpart. Let $g_P(\varphi) = \kappa\varphi_{\min}^{\kappa}\varphi^{-(\kappa+1)}$ and $\Phi_P(\varphi^*) = \kappa(\kappa - \sigma + 1)^{-1}(\varphi_{\min}/\varphi^*)^{\kappa}$ denote the Pareto density and revenue aggregate.

Definition 1 (Density and Tail Deviations)**.** $\rho(\varphi^*) \equiv g(\varphi^*)/g_P(\varphi^*) - 1$ *and* $\psi(\varphi^*) \equiv \Phi(\varphi^*)/\Phi_P(\varphi^*) - 1$.

Economically, ρ captures how much the mass of marginal firms (those at the cutoff) deviates from Pareto, while ψ captures how much the revenue contribution of all inframarginal firms (those above the cutoff) deviates. Under Pareto, direct integration (Appendix A.2) gives $\varphi^* g_P(\varphi^*)/\Phi_P(\varphi^*) = \kappa - (\sigma - 1)$. Substituting $g = g_P(1 + \rho)$ and $\Phi = \Phi_P(1 + \psi)$ into Equation (4):

$$\varepsilon(\varphi^*) = (\sigma - 1) + \frac{\varphi^* g_P(\varphi^*)}{\Phi_P(\varphi^*)} \cdot \frac{1 + \rho(\varphi^*)}{1 + \psi(\varphi^*)} = (\sigma - 1) + [\kappa - (\sigma - 1)] \cdot \frac{1 + \rho(\varphi^*)}{1 + \psi(\varphi^*)}. \tag{6}$$

Subtracting κ and simplifying yields the elasticity departure from Pareto:

$$\varepsilon(\varphi^*) - \kappa = [\kappa - (\sigma - 1)] \cdot \frac{\rho(\varphi^*) - \psi(\varphi^*)}{1 + \psi(\varphi^*)}. \tag{7}$$

Equation (7) reveals that the elasticity departure is governed by the wedge $\rho - \psi$, not by ρ or ψ individually. A distribution with excess mass at the cutoff ($\rho > 0$) but a proportionally heavier tail ($\psi = \rho$) generates no elasticity departure at all. This is economically natural: what distorts the welfare calculation is not whether there are "too many" or "too few" firms relative to Pareto, but whether the density and the revenue-weighted tail are distorted asymmetrically.

Let $[\varphi_L, \varphi_H]$ denote the range of export cutoffs traversed along the equilibrium path induced by the policy shock, analogous to $[z_L, z_H]$ in the EK class.

Assumption 2. $\Phi(\varphi) > 0$ *for all* $\varphi \in [\varphi_L, \varphi_H]$.

Assumption 2 requires a positive mass of firms above the cutoff throughout the equilibrium path. It is violated only if the distribution has finite support and the cutoff reaches the upper bound during the transition.

Proposition 2 (Melitz Bound)**.** *Under the four ACR conditions, Assumption 2, and $\kappa > \sigma - 1$:*

$$|\mathcal{E}_{\text{Melitz}}| \leq \frac{\kappa - (\sigma - 1)}{(\sigma - 1)\,\kappa} \cdot \sup_{\varphi \in [\varphi_L, \varphi_H]} \frac{|\rho(\varphi) - \psi(\varphi)|}{1 + \psi(\varphi)} \cdot |\ln \hat{\lambda}|. \tag{8}$$

See Appendix A.4 for the proof.

Comparing Equations (5) and (8) highlights an important asymmetry between the two model classes. The EK bound loads linearly on the hazard-rate deviation $|\eta|/(\theta + \eta)$. The Melitz bound loads on the asymmetric wedge $|\rho - \psi|/(1 + \psi)$, but with an additional dampening factor $[\kappa - (\sigma - 1)]/[(\sigma - 1)\kappa]$ that vanishes as $\kappa \downarrow \sigma - 1$. Economically, the Melitz elasticity in Equation (4) has a structural floor at $\sigma - 1$ from the CES intensive margin, which compresses the admissible range of ε relative to the EK class where the hazard rate carries no such floor. The Monte Carlo results in Section 5 confirm that, for the same underlying non-Pareto distribution, the EK welfare error is systematically larger than the Melitz one.

Corollary 2 (Melitz Local Sufficiency)**.** *If $\rho(\varphi) = \psi(\varphi)$ for all $\varphi \in [\varphi_L, \varphi_H]$, then $\mathcal{E}_{\text{Melitz}} = 0$.*

As with Corollary 1, the condition is local: the density and tail deviations need only coincide along the segment of the distribution traversed by the policy shock.

3.2 Macro-Observable Reformulation

The micro-level bounds require firm-level distributional data to estimate η, ρ, and ψ. This limits their practical applicability, since the researchers who use the ACR formula typically lack access to firm-level productivity data. We now show that the bounds admit equivalent reformulations in terms of ε alone. This is the key theoretical step: it connects the distributional primitives to an object that is directly estimable from aggregate bilateral trade data, and thereby enables the gravity-based diagnostic developed in Section 4.

Proposition 3 (Macro EK Bound)**.** *Under the conditions of Proposition 1:*

$$|\mathcal{E}_{\text{EK}}| \leq \sup_{z \in [z_L, z_H]} \frac{|\varepsilon(z) - \theta|}{\varepsilon(z)} \cdot |\ln \hat{\lambda}|. \tag{9}$$

Proof. Substitute $\eta(z) = \varepsilon(z) - \theta$ into Equation (5). □

Corollary 3 (Simplified EK Bound)**.** *Under Assumption 1 with $\varepsilon(z) \geq \sigma - 1$:*

$$|\mathcal{E}_{\text{EK}}| \leq \frac{\sup_z |\varepsilon(z) - \theta|}{\sigma - 1} \cdot |\ln \hat{\lambda}|. \tag{10}$$

Proof. From Equation (9), replace $\varepsilon(z)$ in the denominator with its lower bound $\sigma - 1$. □

Corollary 3 brings the EK bound into the same scale as the Melitz bound (Equation (11)), with both denominators proportional to $\sigma - 1$. The key difference is that the Melitz bound carries the additional factor $1/\kappa$, reflecting the dampening from the revenue-weighted aggregate Φ. When Assumption 1 is violated, Corollary 3 does not apply, but the original bound in Proposition 3 remains valid with $\varepsilon(z)$ in the denominator.

Proposition 4 (Macro Melitz Bound)**.** *Under the conditions of Proposition 2:*

$$|\mathcal{E}_{\text{Melitz}}| \leq \frac{1}{(\sigma - 1)\,\kappa} \cdot \sup_{\varphi^*} |\varepsilon(\varphi^*) - \kappa| \cdot |\ln \hat{\lambda}|. \tag{11}$$

Proof. Substitute Equation (7) into Equation (8). □

The practical import of Equations (9) and (11) is that neither η, ρ, nor ψ appears. The bounds depend entirely on the variation of the aggregate trade elasticity ε relative to its benchmark value. Under Fréchet or Pareto, ε is constant, so any observed variation in ε across trade cost levels is diagnostic of distributional departure, and the magnitude of that variation is all a researcher needs to bound the welfare error.

3.3 Rate of Convergence to Sufficiency

The sufficiency conditions ($\eta = 0$ in EK, $\rho = \psi$ in Melitz) may appear restrictive. In fact, they are not knife-edge: many commonly used distributions satisfy them approximately in the upper tail, and the approximation improves as the cutoff moves deeper into the tail. We formalize this using second-order regular variation (De Haan and Ferreira, 2006), which provides a natural framework for quantifying the rate at which a distribution converges to its Fréchet/Pareto limit.

Definition 2 (Second-Order Regular Variation)**.** $\bar{G} \in 2\text{RV}(-\kappa, \beta)$ *with* $\beta \leq 0$ *if there exists* $A(\varphi) \to 0$, $|A| \in \text{RV}(\beta)$, *such that for all* $t > 0$*:*

$$\lim_{\varphi \to \infty} \frac{\bar{G}(t\varphi)/\bar{G}(\varphi) - t^{-\kappa}}{A(\varphi)} = t^{-\kappa} \cdot \frac{t^{\beta} - 1}{\beta}, \tag{12}$$

with the convention $(t^{\beta} - 1)/\beta \to \ln t$ *as* $\beta \to 0$*. The first-order parameter* κ *is the Pareto tail index. The second-order parameter* $\beta \leq 0$ *governs the convergence rate:* β *closer to zero means slower convergence. The auxiliary function* $A(\varphi) \to 0$ *captures the local magnitude of the departure from exact Pareto behavior.*

Lemma 1 (Convergence Rate)**.** *If the technology distribution satisfies* $F_0 \in 2\mathrm{RV}(-\theta, \beta)$ *with auxiliary function* A*, then* $|\eta(z)|/(\theta + \eta(z)) = O(|A(z)|)$ *as* $z \to \infty$*. Analogously, if* $\bar{G} \in 2\mathrm{RV}(-\kappa, \beta)$*, then* $|\rho(\varphi^*) - \psi(\varphi^*)| = O(|A(\varphi^*)|)$ *as* $\varphi^* \to \infty$*.*

See Appendix A.5 for the proof.

Lemma 1 has two practical implications. First, for distributions with slow tail convergence, the ACR error may be non-negligible even at high cutoffs. The leading example is the log-normal distribution, which has $\beta = 0$ and $|A(\varphi)| = O(1/\ln \varphi)$: the sufficiency wedge $|\rho - \psi|$ shrinks only logarithmically, so it persists at empirically relevant cutoffs. This is consistent with the well-documented difficulty of distinguishing log-normal from Pareto tails in firm-size data (Head et al., 2014). Second, for distributions with rapid convergence, the ACR formula becomes reliable even if the distribution is globally non-Pareto. The truncated Pareto ($\beta < 0$, $|A(\varphi)| \sim \varphi^{\beta}$) is the canonical example: the wedge shrinks polynomially, and the ACR error vanishes rapidly as the cutoff increases.

Lemma 1 is stated for $z, \varphi^* \to \infty$, which corresponds to policy shocks that move the marginal cutoff deeper into the tail of the distribution. In the EK and Melitz models, trade liberalization (a reduction in τ) raises φ^*, so the convergence argument applies directly to the NAFTA application in Section 6. For restrictive shocks that lower φ^* into the body of the distribution, the convergence rate may be slower, and the ACR error can persist or grow with the shock magnitude.

4 Gravity-Based Diagnostic

The macro-observable bounds in Equations (9) and (11) require the variation of ε across trade cost levels. We now describe a concrete procedure to estimate this variation from bilateral trade data and evaluate the bound. The structural derivations connecting the trade elasticity formulas in Equations (3) and (4) to gravity are provided in Appendices A.1 and A.2; here we focus on the empirical implementation.

In both model classes, bilateral trade depends on iceberg trade costs τ_{ij}, and the gravity elasticity $-\partial \ln X_{ij}/\partial \ln \tau_{ij} = \varepsilon$ is constant under the distributional assumption. In empirical gravity equations, bilateral distance d_{ij} proxies for trade costs: $\ln \tau_{ij} = \delta \ln d_{ij} + \ln h_{ij}$ for some $\delta > 0$, where h_{ij} collects non-distance barriers absorbed by fixed effects. The standard gravity regression:

$$\ln X_{ij} = S_i + M_j - \varepsilon \ln d_{ij} + \gamma' \mathbf{B}_{ij} + u_{ij} \tag{13}$$

(with exporter and importer fixed effects S_i, M_j, and bilateral controls $\mathbf{B}_{ij}$ including contiguity, common official language, colonial relationship, and regional trade agreement) recovers a constant distance coefficient under the distributional assumption.[3] Under non-standard distributions, the gravity elasticity varies with d_{ij}, inducing a nonlinear relationship between log trade and log distance. This nonlinearity motivates flexible gravity specifications in the spirit of Helpman et al. (2008).

4.1 Procedures

The diagnostic consists of three steps.[4]

Step 1: Pooled gravity. Estimate Equation (13) by Poisson Pseudo Maximum Likelihood (PPML) to obtain the benchmark elasticity $\hat{\varepsilon}_{\text{pool}} \equiv -\hat{\beta}_1$ (Silva and Tenreyro, 2006). This is the constant elasticity that enters the standard ACR formula and serves as the baseline against which departures are measured.

Step 2: Augmented gravity. Re-estimate the gravity equation with a quadratic distance term:

$$X_{ij} = \exp\big(S_i + M_j + \beta_1 \ln d_{ij} + \beta_2 (\ln d_{ij})^2 + \gamma' \mathbf{B}_{ij}\big) \cdot u_{ij}. \tag{14}$$

Under Fréchet/Pareto, $\beta_2 = 0$ and the gravity relationship is exactly log-linear. If $\hat{\beta}_2$ is statistically different from zero, the trade elasticity varies across bilateral pairs:

$$\varepsilon(d_{ij}) = -(\beta_1 + 2\beta_2 \ln d_{ij}). \tag{15}$$

The implied range of elasticity variation across observed distances is:

$$\widehat{\Delta\varepsilon} = 2|\hat{\beta}_2| \cdot (\ln d_{\max} - \ln d_{\min}). \tag{16}$$

The test $H_0\colon \beta_2 = 0$ is thus a direct test of the Fréchet/Pareto distributional assumption, conditional on the four ACR macro-restrictions.

[3] See Head and Mayer (2014) and Yotov et al. (2016) for a comprehensive review of gravity specifications.

[4] An alternative to Step 2 is to partition bilateral pairs into trade cost quantiles and estimate separate elasticities within each bin. We adopt the quadratic specification as the baseline because it uses the full sample efficiently. Any method that reliably identifies elasticity variation can serve the same purpose.

Step 3: Bound evaluation. Substitute $\widehat{\Delta\varepsilon}$ into the macro-observable bounds. For EK, applying Corollary 3:

$$|\hat{\mathcal{E}}_{\text{EK}}| \leq \frac{\widehat{\Delta\varepsilon}}{\sigma - 1} \cdot |\ln \hat{\lambda}|. \tag{17}$$

For Melitz:

$$|\hat{\mathcal{E}}_{\text{Melitz}}| \leq \frac{\widehat{\Delta\varepsilon}}{(\sigma - 1)\,\hat{\varepsilon}_{\text{pool}}} \cdot |\ln \hat{\lambda}|. \tag{18}$$

All inputs are available from Steps 1 and 2 ($\hat{\varepsilon}_{\text{pool}}$, $\hat{\beta}_2$, distance range), the elasticity of substitution σ, and the observed change in the domestic expenditure share $\hat{\lambda}$.[5] If $\hat{\beta}_2$ is insignificant, the researcher concludes that the ACR formula is reliable at the precision afforded by the data. If $\hat{\beta}_2$ is significant, the bound quantifies the worst-case welfare error and the researcher can assess whether it is economically tolerable for the application at hand.

Cross-sectional variation as a bound on the counterfactual path. The theoretical error in Equation (2) is defined along the counterfactual path, whereas the diagnostic estimates elasticity variation across bilateral pairs. The connection rests on the observation that, within the ACR framework, different bilateral pairs face different cutoffs φ^*_{ij}, so the cross-sectional variation in $\varepsilon(d_{ij})$ traces out the same function $\varepsilon(\cdot)$ that governs the counterfactual path, evaluated at different points of its domain. The bound is valid for shocks that move equilibrium cutoffs within the range $[d_{\min}, d_{\max}]$ spanned by observed pairs; for shocks that create relationships at distances outside this range, the bound may understate the true error.

4.2 Identification

A natural question is whether the test $H_0\colon \beta_2 = 0$ has a structural interpretation beyond a statistical specification check. The following result confirms that, within EK and Melitz models satisfying the four ACR macro-restrictions, gravity curvature is equivalent to distributional departure.

[5] Since distance proxies for trade costs through $\ln \tau_{ij} = \delta \ln d_{ij} + \ln h_{ij}$, the gravity coefficients estimate $\varepsilon\delta$ rather than ε. In the Melitz bound (Equation (18)), δ cancels between the numerator $\widehat{\Delta\varepsilon}$ and the denominator $\hat{\varepsilon}_{\text{pool}}$, so the bound is invariant to δ. In the EK bound (Equation (17)), $\widehat{\Delta\varepsilon}$ retains the factor δ, making the bound conservative when $\delta > 1$.

Remark 1 (Local Identification)**.** *Under the four ACR conditions, a continuously differentiable distribution, and the quadratic specification of Equation* (14)*,* $\beta_2 = 0$ *if and only if the distribution is Fréchet (EK) or Pareto (Melitz) on the support spanned by observed bilateral pairs. The argument is straightforward: in the EK case,* $\beta_2 = 0$ *implies* $\varepsilon(z) = \theta$ *for all observed margins, which from Equation* (3) *yields* $d\ln(1 - F_0)/d\ln z = -\theta$ *with unique solution* $F_0(z) = \exp(-z^{-\theta})$*. The Melitz case is analogous (see Appendix A.6).*

If the macro-restrictions themselves are violated, $\beta_2 \neq 0$ still implies that the ACR formula with constant ε is unreliable, and the computed bound provides a lower bound on the total welfare error. The aggregate gravity curvature detected by $\hat{\beta}_2$ also has a micro-level counterpart: using matched French–Chinese customs data, Bas et al. (2017) document that the bilateral trade elasticity varies systematically across country pairs precisely because firm-level productivity departs from Pareto. Our macro-observable diagnostic captures, at the cross-sectional level, what their firm-level decomposition identifies directly.

Limitations. Three caveats apply. First, $\widehat{\Delta\varepsilon}$ is estimated from observed bilateral distances; if the distribution exhibits larger departures at cutoffs outside this range, the empirical bound may understate the theoretical supremum. Our distance range (174–19,812 km) spans nearly the full range of feasible bilateral distances, mitigating this concern. Second, the quadratic specification may fail to detect localized departures; the quantile-based alternative in footnote 4 provides a nonparametric check at the cost of reduced precision. Third, if observed curvature partly reflects non-distributional sources of trade-cost nonlinearity, the reported bound overstates the true ACR error, making the diagnostic conservative rather than less so.

5 Monte Carlo Validation

We simulate symmetric 30-country economies under four data-generating processes (DGPs), each representing a distinct pattern of distributional departure from the constant-elasticity benchmark. In all cases we set $\sigma = 4$ and draw bilateral trade costs uniformly from $[1.1, 3.0]$. Since the EK and Melitz models generate different trade flow structures from the same underlying distribution, we run the diagnostic separately for each model class: EK trade flows are proportional to the survival function $1 - F_0(z)$, while Melitz trade flows are proportional to the revenue-weighted aggregate $\Phi(\varphi^*)$. The four DGPs are:

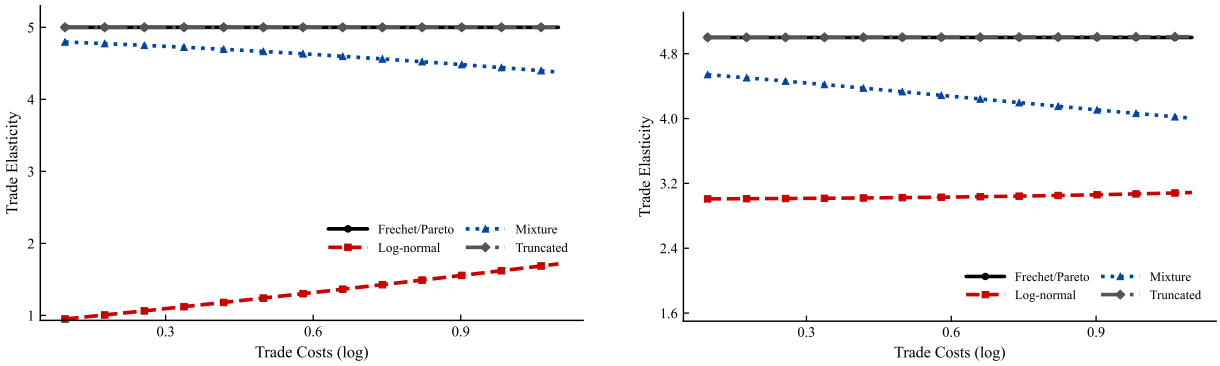
(i) *Fréchet/Pareto* ($\theta = \kappa = 5$). The exact benchmark: the trade elasticity is constant at 5, and $\hat{\beta}_2 = 0$ by construction. Any nonzero error or bound in this case reflects only numerical precision.

(ii) *Log-normal* ($\mu = 0$, $\sigma_{\ln} = 1$). A distribution with no power-law tail. The hazard rate varies continuously across the support, so the EK elasticity departs substantially from any constant. The Melitz elasticity is dampened by the $\sigma - 1$ floor but still varies. This DGP represents the empirically relevant scenario in which the Pareto approximation is poor throughout the distribution.

(iii) *Pareto mixture* ($\kappa_1 = 5$, $\kappa_2 = 3.5$, $\pi = 0.9$). A 90–10 mixture of two Pareto distributions with different shape parameters. The heavier-tailed component ($\kappa_2 = 3.5$) contaminates the benchmark ($\kappa_1 = 5$), creating localized density bunching that the diagnostic should detect. This DGP captures scenarios where a subpopulation of firms (e.g., multinationals) follows a different productivity distribution, and closely parallels the Pareto–log-normal mixture documented by Nigai (2017).

(iv) *Truncated Pareto* ($\kappa = 5$, $\bar{\varphi} = 50$). A Pareto distribution with an upper bound on productivity. The truncation introduces a second-order departure that shrinks polynomially in the cutoff (Lemma 1), so the ACR formula should remain accurate. This DGP tests whether the diagnostic correctly identifies benign departures.

For each DGP, we compute population-level trade flows and apply the three-step gravity diagnostic of Section 4.



(a) EK class (b) Melitz class

Figure 1: Trade Elasticity Profiles across DGPs

Figure 1 plots the gravity elasticity profile for each DGP. Under Fréchet/Pareto, ε is flat in both panels. The key difference between the two model classes is visible under

the log-normal DGP: in the EK class (panel a), the elasticity drops sharply across the trade cost range, whereas in the Melitz class (panel b) the variation is compressed by the structural floor at $\sigma - 1$. Under the mixture, ε varies substantially in both classes. The truncated Pareto is indistinguishable from the benchmark.

Table 1 reports the curvature coefficient $\hat{\beta}_2$ and implied elasticity variation for both model classes. Under Fréchet/Pareto, $\hat{\beta}_2 = 0$ exactly in both cases. The most striking difference appears under the log-normal DGP: the EK diagnostic detects substantial curvature ($\hat{\beta}_2 = -0.384$, $\widehat{\Delta\varepsilon} = 0.768$), whereas the Melitz diagnostic finds only modest curvature ($\hat{\beta}_2 = -0.038$, $\widehat{\Delta\varepsilon} = 0.077$). In the Melitz elasticity (Equation (4)), the $\sigma - 1$ floor arises automatically from the additive structure, compressing the range of elasticity variation relative to the EK class, where the hazard rate carries no such inherent bound.

Table 1: Monte Carlo: Gravity Diagnostic

	EK class			Melitz class		
DGP	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	$\widehat{\Delta\varepsilon}$	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	$\widehat{\Delta\varepsilon}$
Fréchet/Pareto	5.000	0.000	0.000	5.000	0.000	0.000
Log-normal	1.340	−0.384	0.768	3.036	−0.038	0.077
Mixture ($\pi = 0.9$)	4.606	0.213	0.426	4.260	0.276	0.552
Truncated ($\bar{\varphi} = 50$)	5.000	0.000	0.000	5.004	−0.004	0.008

Notes: Population-level gravity diagnostic on simulated trade data. EK trade flows use the survival function $1 - F_0(z)$; Melitz trade flows use the revenue-weighted aggregate $\Phi(\varphi^*)$. $\hat{\beta}_2 = 0$ under Fréchet/Pareto in both classes. Parameters: $\sigma = 4$, $\theta = \kappa = 5$, $N = 30$, $\tau \in [1.1, 3.0]$.

Table 2 compares true ACR errors to the diagnostic bounds for both model classes. All bounds hold across all DGPs and shock sizes. Two patterns are noteworthy. First, the EK class is far more vulnerable to distributional departures than the Melitz class. Under the log-normal DGP, the EK error reaches 13.6% of real income for a 40% trade cost reduction, compared to just 0.21% in the Melitz class. Although Assumption 1 imposes a structural floor at $\sigma - 1$ on both model classes, the EK bound (Equation (17)) scales as $\widehat{\Delta\varepsilon}/(\sigma - 1)$, whereas the Melitz bound (Equation (18)) carries the additional dampening factor $1/\hat{\varepsilon}_{\text{pool}}$. Under the log-normal DGP, the EK diagnostic detects much larger elasticity variation ($\widehat{\Delta\varepsilon} = 0.768$) than the Melitz diagnostic ($\widehat{\Delta\varepsilon} = 0.077$), yielding a bound ratio of roughly 30:1. Note that the log-normal DGP violates Assumption 1 in the EK class ($\varepsilon_{\min} = 0.94 < \sigma - 1 = 3$), so the EK bound for this DGP is computed from Proposition 3 rather than Corollary 3; it nonetheless holds in all cases.

Second, the bounds are conservative but informative. Under the mixture DGP, the

Melitz bound at a 40% reduction is 7.58% versus a true error of 1.07%, a ratio of roughly 7:1. This conservatism is inherent in any supremum-based bound, but the bound nonetheless correctly identifies the mixture as the most problematic DGP and provides a quantitative ceiling that a researcher can evaluate against the magnitude of the welfare calculation.

Table 2: Monte Carlo: ACR Welfare Error and Diagnostic Bounds

		EK class			Melitz class		
DGP		10%	25%	40%	10%	25%	40%
Fréchet/Pareto	Error	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	Bound	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Log-normal	Error	0.11%	2.85%	13.61%	0.01%	0.07%	0.21%
	Bound[a]	6.61%	18.83%	34.71%	0.22%	0.62%	1.15%
Mixture ($\pi = 0.9$)	Error	<0.01%	0.17%	0.70%	0.01%	0.23%	1.07%
	Bound	4.01%	12.50%	24.91%	1.22%	3.80%	7.58%
Truncated ($\bar{\varphi} = 50$)	Error	<0.01%	<0.01%	<0.01%	0.00%	0.00%	0.01%
	Bound	<0.01%	<0.01%	<0.01%	0.02%	0.05%	0.10%

Notes: Error is $|\ln \hat{W}_{\text{exact}} - \ln \hat{W}_{\text{ACR}}|$ as a percentage of real income. EK bound from Equation (17); Melitz bound from Equation (18). Each model class uses its own diagnostic ($\widehat{\Delta\varepsilon}$, $\hat{\varepsilon}_{\text{pool}}$) estimated from the corresponding trade flow structure. Trade cost reduced from 2.0 to 1.8/1.5/1.2. Parameters: $\sigma = 4$, $\theta = \kappa = 5$, $N = 30$. [a]For the log-normal EK bound, Assumption 1 is violated; the bound is computed from Proposition 3.

6 Application: NAFTA Welfare Effects

We apply the diagnostic to the welfare calculation of CP, the leading quantitative evaluation of a regional trade agreement. CP quantify the welfare effects of NAFTA using a multi-country, multi-sector Ricardian model with intermediate goods and input-output linkages, extending EK to 20 tradable and 20 non-tradable sectors across 31 countries (the NAFTA members plus 27 trading partners and a rest-of-world aggregate). Their welfare estimates are the standard benchmark for North American trade integration, and their modeling approach has been adopted in subsequent analyses of trade agreements worldwide.

CP's model assumes Fréchet productivity distributions in each tradable sector, and the sectoral ACR formula enters their welfare decomposition as a building block. From

their Equation (15), the change in real wages decomposes as:

$$\ln \frac{\hat{w}_n}{\hat{P}_n} = -\sum_{j=1}^{J} \frac{\alpha_n^j}{\theta^j} \ln \hat{\pi}_{nn}^j + (\text{intermediate goods and sectoral linkage terms}), \tag{19}$$

where α_n^j is sector j's share in final demand, θ^j is the sectoral trade elasticity (Fréchet dispersion parameter), and $\hat{\pi}_{nn}^j$ is the change in the domestic expenditure share. The first term is a weighted sum of sectoral ACR contributions. If the Fréchet assumption is violated in sector j, the trade elasticity is no longer constant at θ^j and that sector's contribution acquires an approximation error bounded by our diagnostic. The remaining terms capture intermediate goods linkages and are unaffected by the within-sector distributional assumption.

Two features make CP an ideal application. First, because their model belongs to the EK class, the constant profit share condition (iv) holds automatically regardless of the distributional assumption (Section 2), so the conceptual concern raised by Melitz and Redding (2015) about the entry margin does not apply. Second, CP estimate sector-specific trade elasticities θ^j from tariff variation rather than from gravity, so the structural parameters are identified independently of the distance-cost mapping δ discussed in footnote 5.

6.1 Data and Implementation

We use two inputs. The first is CP's replication package, which provides bilateral trade flows across 20 tradable sectors for 31 countries in 1993 (the NAFTA baseline year), sector-specific trade elasticities θ^j, final demand shares α_n^j, and input-output coefficients. We run CP's counterfactual code to compute the NAFTA-only scenario: bilateral tariffs between Mexico, Canada, and the U.S. are reduced from 1993 to 2005 levels, while tariffs with all other countries are held fixed. This yields the counterfactual domestic share changes $\hat{\pi}_{nn}^j$ for each sector and country under general equilibrium wage and price adjustment.

The second input is geodesic distances and bilateral controls (contiguity, common official language, colonial relationship) from the CEPII Gravity database (Conte et al., 2022). CP's replication package does not include bilateral distances because their model solves for counterfactual outcomes using tariff data and hat algebra, without requiring a gravity regression. We match CEPII distances to CP's 30 countries (excluding the rest-of-world aggregate, for which no meaningful bilateral distance exists), yielding 870

bilateral pairs.

For each of the 20 tradable sectors, we estimate the quadratic gravity specification (Equation (14)) by PPML with exporter and importer fixed effects using the 1993 bilateral trade flows and CEPII distances. This yields a sectoral elasticity variation $\widehat{\Delta\varepsilon}_j$ for each sector, estimated from the same country sample and base year as CP's model. We then compute the welfare-weighted EK bound for each sector and country:

$$\text{Bound}_n = \sum_{j=1}^{J} \frac{\alpha_n^j}{\theta^j} \cdot \frac{\widehat{\Delta\varepsilon}_j}{\sigma - 1} \cdot |\Delta \ln \hat{\pi}_{nn}^j|, \tag{20}$$

evaluated at $\sigma = 4$. The full sectoral gravity diagnostic and country-level bounds are reported in Appendix B; a complementary exercise at the finer HS 2-digit level using 2019 BACI trade data is reported in Appendix C.

6.2 Results

Table 3: ACR error bounds on NAFTA welfare calculation

Country	CP welfare (pp)	EK bound (pp)	Largest sector
Mexico	1.31	0.10	Auto (0.026)
Canada	−0.06	0.07	Auto (0.022)
U.S.	0.08	0.01	Other Manuf (0.002)

Notes: CP welfare from Caliendo and Parro (2015, Table 2), NAFTA-only counterfactual without trade deficits. EK bound from Equation (20), evaluated at $\sigma = 4$, using counterfactual $|\Delta \ln \hat{\pi}_{nn}^j|$ from CP's general equilibrium solution. Sectoral $\widehat{\Delta\varepsilon}$ estimated by PPML from CP's 1993 bilateral trade data with CEPII distances. Largest sector reports the single sector with the largest welfare-weighted bound contribution.

Table 3 reports the results. Of the 20 tradable sectors, 6 reject $H_0\colon \beta_2 = 0$ at the 5% level: mining, paper, chemicals, basic metals, auto, and other manufacturing (see Table B1 in Appendix B for the full sectoral diagnostic). CP report that NAFTA increased Mexico's welfare by 1.31 percentage points.[6] The aggregate EK error bound from the Fréchet assumption is 0.10 percentage points, implying that Mexico's welfare gain lies between 1.21% and 1.41%. The bound for the U.S. is 0.01 pp (welfare gain 0.08%), and for Canada 0.07 pp (welfare change −0.06%).

[6]The bound is on $|\mathcal{E}|$, the absolute value of the approximation error $\mathcal{E} \equiv \ln \hat{W}_{\text{exact}} - \ln \hat{W}_{\text{ACR}}$. Since $\mathcal{E}$ can be positive or negative, the ACR formula may either overstate or understate the true welfare change, and the interval is two-sided.

The sectoral decomposition reveals that the bound is driven by auto (0.026 pp), other manufacturing (0.020 pp), and petroleum (0.014 pp). The auto sector combines a large counterfactual domestic share change ($|\Delta \ln \hat{\pi}_{nn}| = 0.13$ for Mexico) with significant gravity curvature ($\hat{\beta}_2 = -0.15$, $p = 0.01$), reflecting the highly concentrated structure of North American auto production and the large NAFTA tariff reductions in this sector. Sectors with small $\widehat{\Delta\varepsilon}$ (e.g., machinery, $p = 0.61$) or small welfare weight (e.g., wood, $\alpha = 0$) contribute negligibly regardless of the shock magnitude.

This sectoral pattern points to a broader empirical regularity. The distributional assumption matters most where three conditions coincide: the policy shock is large (so the equilibrium path traverses a wide range of cutoffs), the sector receives substantial weight in welfare (so the sectoral error translates into a meaningful aggregate error), and the sector exhibits significant within-sector productivity heterogeneity captured by gravity curvature. The auto sector in NAFTA satisfies all three; sectors that fail any one of them, even if they violate Fréchet, contribute little to the aggregate bound.

To put the bound in context, CP's own robustness analysis reports welfare estimates ranging from 0.11% to 1.90% for Mexico across alternative trade elasticity parameterizations (their Table 3), a range of 1.79 pp. The 0.10 pp distributional bound is an order of magnitude smaller. Simonovska and Waugh (2014) similarly document that alternative estimation methods yield trade elasticities ranging from 4 to 8, nearly doubling the implied welfare gains. Uncertainty in the trade elasticity estimate itself thus dominates uncertainty from the Fréchet assumption.

7 Conclusion

This paper provides macro-observable upper bounds on the ACR formula's approximation error when the Fréchet or Pareto assumption is violated, computable from standard gravity data without firm-level microdata.

Applied to CP's NAFTA calculation, the aggregate error bound for Mexico is 0.10 pp against a reported welfare gain of 1.31%, and 0.01 pp for the U.S. The error is concentrated in sectors with large counterfactual trade expansion and detectable gravity curvature. The Fréchet assumption does not overturn the qualitative conclusion. As a practical guideline, researchers should compare the computed bound to the welfare effect they report and assess whether the distributional assumption is consequential for the policy conclusion at hand.

Several extensions merit investigation. First, violations of the macro-restrictions

themselves, most notably variable markups (e.g., Atkeson and Burstein, 2008; Arkolakis et al., 2019; Firooz and Heins, 2026), require alternative sufficient statistics and a different analytical framework. Second, applying the diagnostic to more recent episodes such as the U.S.-China trade war would test the bounds at larger shock magnitudes. Third, the sectoral patterns suggest a link between product characteristics (homogeneity, perishability, regulatory barriers) and distributional form that deserves theoretical formalization.

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A Proofs and Derivations

A.1 Derivation of the EK Trade Elasticity

In the EK model, country j sources each good ω from the lowest-cost supplier. Country i offers good ω in market j at price:

$$p_{ij}(\omega) = \frac{c_i \, \tau_{ij}}{z_i(\omega)}, \tag{A1}$$

where c_i is the unit input cost, $\tau_{ij} \geq 1$ is the iceberg trade cost, and $z_i(\omega)$ is an i.i.d. efficiency draw from F_0. Country j imports from the cheapest source. The share of goods for which i is the lowest-cost supplier determines bilateral trade. Conditional on the technology parameters and wages of all other countries (which are absorbed into the exporter and importer fixed effects in the gravity equation), bilateral trade satisfies:

$$X_{ij} \propto 1 - F_0(\tilde{z}_{ij}), \tag{A2}$$

where $\tilde{z}_{ij}$ is the marginal technology draw for pair (i, j), with $\tilde{z}_{ij}$ increasing in τ_{ij}. The gravity elasticity is therefore:

$$-\frac{\partial \ln X_{ij}}{\partial \ln \tau_{ij}} = \frac{\tilde{z}_{ij} \, f_0(\tilde{z}_{ij})}{1 - F_0(\tilde{z}_{ij})} = \varepsilon(\tilde{z}_{ij}), \tag{A3}$$

which is Equation (3). Under Fréchet, $F_0(z) = \exp(-z^{-\theta})$, so:

$$f_0(z) = \theta \, z^{-(\theta+1)} \exp(-z^{-\theta}), \tag{A4}$$

$$1 - F_0(z) = 1 - \exp(-z^{-\theta}). \tag{A5}$$

Direct computation gives:

$$\varepsilon(z) = \frac{z \cdot \theta \, z^{-(\theta+1)} \exp(-z^{-\theta})}{1 - \exp(-z^{-\theta})} = \frac{\theta \, z^{-\theta} \exp(-z^{-\theta})}{1 - \exp(-z^{-\theta})}. \tag{A6}$$

Let $u \equiv z^{-\theta}$. Then $\varepsilon = \theta u e^{-u}/(1 - e^{-u})$. Using $1 - e^{-u} = u e^{-u} \sum_{k=0}^{\infty} u^k/(k+1)!$ shows that $\varepsilon = \theta$ exactly when $u \to 0$ (i.e., $z \to \infty$), which is the relevant range for marginal exporters. More precisely, $\varepsilon(z) = \theta + O(z^{-\theta})$, confirming that the elasticity is exactly constant in the Fréchet limit and approximately constant for large z.

A.2 Derivation of the Melitz Trade Elasticity

The derivation here follows Chaney (2008), who provides the canonical mapping from the Melitz (2003) model to a gravity equation with constant elasticity under Pareto productivity. In the Melitz model with CES preferences (elasticity $\sigma > 1$), a firm with productivity φ in country i earns revenue from sales to country j equal to:

$$r_{ij}(\varphi) = A_j \left(\frac{\tau_{ij}\, w_i}{\varphi} \right)^{1-\sigma}, \tag{A7}$$

where A_j is a destination-specific demand shifter. The firm exports to j if and only if variable profits cover the fixed export cost:

$$\frac{r_{ij}(\varphi)}{\sigma} \geq w_i\, f_{ij}. \tag{A8}$$

This defines the zero-profit cutoff:

$$\varphi^*_{ij} = \left(\frac{\sigma\, w_i\, f_{ij}}{A_j} \right)^{\frac{1}{\sigma-1}} \tau_{ij}\, w_i, \tag{A9}$$

which is proportional to τ_{ij}. Aggregate bilateral trade from i to j is:

$$\begin{aligned} X_{ij} &= M_i \int_{\varphi^*_{ij}}^{\infty} r_{ij}(\varphi)\, g(\varphi)\, d\varphi \\ &= M_i\, A_j\, (\tau_{ij}\, w_i)^{1-\sigma}\, (\varphi^*_{ij})^{\sigma-1}\, \Phi(\varphi^*_{ij}), \end{aligned} \tag{A10}$$

where

$$\Phi(\varphi^*) \equiv \int_{\varphi^*}^{\infty} \left(\frac{\varphi}{\varphi^*} \right)^{\sigma-1} g(\varphi)\, d\varphi \tag{A11}$$

collects the revenue contribution of all firms above the cutoff. Since $\varphi^*_{ij} \propto \tau_{ij}$, the gravity elasticity requires differentiating $\ln X_{ij}$ with respect to $\ln \tau_{ij}$. The key step is the derivative

of Φ. Applying the Leibniz rule:

$$\begin{aligned}\frac{d\Phi}{d\varphi^*} &= -g(\varphi^*)\cdot 1 + \int_{\varphi^*}^{\infty} \frac{\partial}{\partial\varphi^*}\left(\frac{\varphi}{\varphi^*}\right)^{\sigma-1} g(\varphi)\,d\varphi \\ &= -g(\varphi^*) - \frac{\sigma-1}{\varphi^*}\int_{\varphi^*}^{\infty}\left(\frac{\varphi}{\varphi^*}\right)^{\sigma-1} g(\varphi)\,d\varphi \\ &= -g(\varphi^*) - \frac{\sigma-1}{\varphi^*}\,\Phi(\varphi^*). \end{aligned} \tag{A12}$$

Converting to log derivatives:

$$\frac{d\ln\Phi}{d\ln\varphi^*} = \frac{\varphi^*}{\Phi}\cdot\frac{d\Phi}{d\varphi^*} = -\frac{\varphi^*\,g(\varphi^*)}{\Phi(\varphi^*)} - (\sigma-1). \tag{A13}$$

The gravity elasticity is therefore:

$$-\frac{\partial\ln X_{ij}}{\partial\ln\tau_{ij}} = (\sigma-1) - \frac{d\ln\Phi}{d\ln\varphi^*} = (\sigma-1) + \frac{\varphi^*\,g(\varphi^*)}{\Phi(\varphi^*)}, \tag{A14}$$

which is Equation (4). Under Pareto, $g(\varphi) = \kappa\,\varphi_{\min}^{\kappa}\,\varphi^{-(\kappa+1)}$, and direct integration gives:

$$\Phi(\varphi^*) = \frac{\kappa}{\kappa-(\sigma-1)}\left(\frac{\varphi_{\min}}{\varphi^*}\right)^{\kappa}, \tag{A15}$$

$$\frac{\varphi^*\,g(\varphi^*)}{\Phi(\varphi^*)} = \kappa-(\sigma-1), \tag{A16}$$

so $\varepsilon(\varphi^*) = (\sigma-1) + [\kappa-(\sigma-1)] = \kappa$, which is constant.

A.3 Proof of Proposition 1

Proof. From Equation (2), the absolute error satisfies:

$$|\mathcal{E}_{\mathrm{EK}}| = \left|\int_{\ln\lambda}^{\ln\lambda'} \frac{\varepsilon(x)-\theta}{\theta\cdot\varepsilon(x)}\,dx\right| \le \int_{\ln\lambda}^{\ln\lambda'} \frac{|\varepsilon(x)-\theta|}{\theta\cdot\varepsilon(x)}\,dx. \tag{A17}$$

Substituting $\eta(z) = \varepsilon(z) - \theta$, the integrand becomes:

$$\frac{|\eta(z)|}{\theta\cdot(\theta+\eta(z))}. \tag{A18}$$

Since $\theta \geq 1$ by Assumption 1, we have:

$$\frac{|\eta(z)|}{\theta \cdot (\theta + \eta(z))} \leq \frac{|\eta(z)|}{\theta + \eta(z)}. \tag{A19}$$

Taking the supremum over $z \in [z_L, z_H]$ and evaluating the remaining integral:

$$|\mathcal{E}_{\text{EK}}| \leq \sup_{x \in [\ln \lambda, \ln \lambda']} \frac{|\eta(x)|}{\theta + \eta(x)} \cdot \left| \int_{\ln \lambda}^{\ln \lambda'} dx \right| = \sup_{z \in [z_L, z_H]} \frac{|\eta(z)|}{\theta + \eta(z)} \cdot |\ln \hat{\lambda}|. \tag{A20}$$

□

A.4 Proof of Proposition 2

Proof. From Equation (2), the absolute error satisfies:

$$|\mathcal{E}_{\text{Melitz}}| \leq \int_{\ln \lambda}^{\ln \lambda'} \frac{|\varepsilon(x) - \kappa|}{\kappa \cdot \varepsilon(x)} \, dx. \tag{A21}$$

In any ACR-class Melitz equilibrium, the zero-profit cutoff φ^* is a strictly monotone function of the iceberg trade cost τ, and τ is in strictly monotone correspondence with the domestic share λ along the equilibrium path (lower trade costs raise φ^* and lower λ). Hence $x = \ln \lambda$ and φ^* are in monotone correspondence on $[\ln \lambda, \ln \lambda']$, so the supremum of the integrand over x equals that over $\varphi \in [\varphi_L, \varphi_H]$. Substituting the wedge relation from Equation (7):

$$\frac{|\varepsilon - \kappa|}{\kappa \cdot \varepsilon} = \frac{[\kappa - (\sigma - 1)]}{\kappa \cdot \varepsilon} \cdot \frac{|\rho - \psi|}{1 + \psi}. \tag{A22}$$

From Equation (4) and Assumption 2, $\varepsilon(\varphi^*) \geq \sigma - 1 > 0$, so:

$$\frac{[\kappa - (\sigma - 1)]}{\kappa \cdot \varepsilon} \leq \frac{\kappa - (\sigma - 1)}{(\sigma - 1)\,\kappa}. \tag{A23}$$

Combining and taking the supremum over $\varphi \in [\varphi_L, \varphi_H]$:

$$|\mathcal{E}_{\text{Melitz}}| \leq \frac{\kappa - (\sigma - 1)}{(\sigma - 1)\,\kappa} \cdot \sup_{\varphi \in [\varphi_L, \varphi_H]} \frac{|\rho(\varphi) - \psi(\varphi)|}{1 + \psi(\varphi)} \cdot |\ln \hat{\lambda}|. \tag{A24}$$

□

A.5 Proof of Lemma 1

Proof. EK class. Under $F_0 \in 2\mathrm{RV}(-\theta, \beta)$, the survival function admits the representation:

$$1 - F_0(tz) = \big(1 - F_0(z)\big) \cdot t^{-\theta} \left[1 + A(z)\,\frac{t^{\beta} - 1}{\beta} + o\big(A(z)\big)\right], \qquad t > 0. \tag{A25}$$

Differentiating both sides with respect to t and evaluating at $t = 1$, the t-derivative of the bracket equals $A(z) + o(A(z))$ (both $t^{-\theta}$ and $(t^{\beta} - 1)/\beta$ contribute through the product rule, and the latter's derivative at $t = 1$ equals 1):

$$-z\, f_0(z) = \big(1 - F_0(z)\big) \cdot \big[-\theta + A(z) + o(A(z))\big]. \tag{A26}$$

Rearranging and dividing by $1 - F_0(z)$:

$$\varepsilon(z) = \frac{z\, f_0(z)}{1 - F_0(z)} = \theta - A(z) + o(A(z)). \tag{A27}$$

Hence $\eta(z) = \varepsilon(z) - \theta = -A(z) + o(A(z))$, and since $\theta + \eta(z) \to \theta > 0$ (using $A(z) \to 0$):

$$\frac{|\eta(z)|}{\theta + \eta(z)} = O\big(|A(z)|\big) \quad \text{as } z \to \infty. \tag{A28}$$

Melitz class. Under $\bar{G} \in 2\mathrm{RV}(-\kappa, \beta)$ with $\beta < 0$, the same argument applied to the survival function yields the density expansion:

$$\frac{g(\varphi)}{g_P(\varphi)} = 1 + \frac{\kappa - \beta}{\kappa\,\beta}\, A(\varphi) + o\big(A(\varphi)\big), \tag{A29}$$

where $g_P(\varphi) = \kappa\, \varphi_{\min}^{\kappa}\, \varphi^{-(\kappa+1)}$ is the Pareto density. The coefficient $(\kappa - \beta)/(\kappa\beta)$ collects two contributions: the $-A(z)$ term carried over from the survival-function expansion, and the additional $A(\varphi)/\kappa$ term arising from differentiating A itself. Consequently,

$$\rho(\varphi) = \frac{g(\varphi)}{g_P(\varphi)} - 1 = \frac{\kappa - \beta}{\kappa\,\beta}\, A(\varphi) + o\big(A(\varphi)\big). \tag{A30}$$

For the revenue-weighted aggregate, substitute Equation (A29) into $\Phi(\varphi^*)$ (Equation (A11)) and split into Pareto and perturbation components. Using $A(\varphi) = a\varphi^{\beta}(1 + o(1))$ for some constant a (by regular variation of $|A|$ with index β) and integrating the

leading-order term:

$$\int_{\varphi^*}^{\infty} \left(\frac{\varphi}{\varphi^*}\right)^{\sigma-1} g_P(\varphi)\, A(\varphi)\, d\varphi = \frac{(\kappa-\sigma+1)}{\kappa-\sigma+1-\beta}\, A(\varphi^*)\, \Phi_P(\varphi^*)\big(1+o(1)\big), \tag{A31}$$

which requires $\kappa-\sigma+1-\beta>0$ for convergence (satisfied whenever $\kappa>\sigma-1$ and $\beta\le 0$). Combining with the Pareto part of g:

$$\psi(\varphi^*) = \frac{\Phi(\varphi^*)}{\Phi_P(\varphi^*)} - 1 = \frac{(\kappa-\beta)(\kappa-\sigma+1)}{\kappa\,\beta\,(\kappa-\sigma+1-\beta)}\, A(\varphi^*) + o\big(A(\varphi^*)\big). \tag{A32}$$

The sufficiency wedge then simplifies:

$$\begin{aligned}\rho(\varphi^*) - \psi(\varphi^*) &= \frac{\kappa-\beta}{\kappa\,\beta}\, A(\varphi^*)\left[1 - \frac{\kappa-\sigma+1}{\kappa-\sigma+1-\beta}\right] + o(A)\\ &= -\frac{\kappa-\beta}{\kappa\,(\kappa-\sigma+1-\beta)}\, A(\varphi^*) + o\big(A(\varphi^*)\big).\end{aligned} \tag{A33}$$

The leading coefficient is finite and bounded (it depends on κ,σ,β but not on φ^*), so $|\rho(\varphi^*)-\psi(\varphi^*)| = O\big(|A(\varphi^*)|\big)$, and since $A(\varphi^*)\to 0$ as $\varphi^*\to\infty$, the wedge vanishes at rate $|A(\varphi^*)|$. □

A.6 Proof of Remark 1

Proof. EK class. The "if" direction is immediate: under Fréchet, $\varepsilon(z)=\theta$ for all z, so gravity is log-linear and $\beta_2=0$. For the "only if" direction, suppose $\beta_2=0$. Then $\varepsilon(z)=\theta$ for all margins spanned by observed bilateral pairs. From Equation (3):

$$\frac{z\, f_0(z)}{1-F_0(z)} = \theta \quad\Longrightarrow\quad \frac{d\ln(1-F_0)}{d\ln z} = -\theta. \tag{A34}$$

This is a first-order linear equation in $\ln(1-F_0)$ with unique solution:

$$1-F_0(z) = C\, z^{-\theta} \tag{A35}$$

for some constant $C>0$, which is the Fréchet survival function.

Melitz class. The "if" direction is again immediate. For the "only if" direction, $\beta_2=0$

implies $\varepsilon(\varphi^*) = \kappa$ for all observed cutoffs. From Equation (4):

$$\frac{\varphi^* g(\varphi^*)}{\Phi(\varphi^*)} = \kappa - (\sigma - 1) \quad \Longrightarrow \quad \frac{d \ln \Phi}{d \ln \varphi^*} = -\kappa. \tag{A36}$$

The unique solution is $\Phi(\varphi^*) = C\,(\varphi^*)^{-\kappa}$, which implies $\bar{G}(\varphi) \propto \varphi^{-\kappa}$. □

B Full Results of NAFTA application

Table B1 reports the ISIC-level gravity diagnostic estimated from CP's 1993 bilateral trade data. Table B2 reports the sectoral error bounds for all three NAFTA members.

Table B1: ISIC Sectoral Gravity Diagnostic (CP Country Sample, 1993)

Sector	θ^j	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	s.e.	p	$\widehat{\Delta\varepsilon}$
Agriculture	9.11	0.848	−0.038	(0.055)	0.484	0.345
Mining	13.53	1.034	−0.332	(0.061)	<0.001	2.993
Food	2.62	0.821	−0.039	(0.041)	0.347	0.348
Textile	8.10	0.786	−0.052	(0.062)	0.401	0.470
Wood	11.50	0.928	−0.029	(0.069)	0.677	0.261
Paper	16.52	0.811	−0.121	(0.037)	0.001	1.091
Petroleum	64.85	1.221	0.093	(0.073)	0.207	0.836
Chemicals	3.13	0.756	−0.058	(0.029)	0.044	0.523
Plastic	1.67	0.873	−0.091	(0.048)	0.057	0.821
Minerals	2.41	0.855	−0.061	(0.034)	0.076	0.551
Basic metals	3.28	0.879	−0.138	(0.035)	<0.001	1.250
Metal products	6.99	0.777	−0.024	(0.042)	0.569	0.217
Machinery nec	1.45	0.608	−0.018	(0.036)	0.607	0.166
Office	12.95	0.663	0.031	(0.056)	0.588	0.276
Electrical	12.91	0.732	−0.073	(0.042)	0.082	0.656
Communication	3.95	0.556	0.029	(0.045)	0.514	0.264
Medical	8.71	0.475	−0.022	(0.031)	0.493	0.195
Auto	8.22	1.050	−0.152	(0.060)	0.012	1.370
Other Transport	8.22	0.129	0.021	(0.060)	0.729	0.189
Other Manuf	3.98	0.550	0.169	(0.057)	0.003	1.522

Notes: PPML with exporter and importer fixed effects, controlling for contiguity, common official language, and colonial relationship; robust standard errors. θ^j: CP's trade elasticity estimates (Table 1 in CP, 99% sample). $\widehat{\Delta\varepsilon} = 2|\hat{\beta}_2|(\ln d_{\max} - \ln d_{\min})$. Sample: 30 countries, 870 pairs. Bilateral trade from CP's replication package (1993); distances from CEPII.

Table B2: Sectoral EK Error Bounds (CP NAFTA Counterfactual)

		Mexico		Canada		U.S.	
Sector	$\widehat{\Delta\varepsilon}$	$\lvert\Delta \ln \hat{\pi}\rvert$	Bound	$\lvert\Delta \ln \hat{\pi}\rvert$	Bound	$\lvert\Delta \ln \hat{\pi}\rvert$	Bound
Agriculture	0.345	0.060	0.004	0.002	0.000	0.005	0.000
Mining	2.993	0.020	0.000	0.006	0.001	0.010	0.000
Food	0.348	0.013	0.005	0.004	0.001	0.001	0.000
Textile	0.470	0.277	0.011	0.118	0.004	0.032	0.001
Wood	0.261	0.682	0.000	0.009	0.000	0.011	0.000
Paper	1.091	0.301	0.000	0.003	0.000	0.003	0.000
Petroleum	0.836	2.217	0.014	0.224	0.001	0.042	0.000
Chemicals	0.523	0.052	0.004	0.004	0.000	0.002	0.000
Plastic	0.821	0.084	0.004	0.024	0.002	0.002	0.000
Minerals	0.551	0.023	0.002	0.013	0.000	0.003	0.000
Basic metals	1.250	0.060	0.000	0.012	0.000	0.005	0.000
Metal products	0.217	0.284	0.002	0.058	0.000	0.004	0.000
Machinery nec	0.166	0.074	0.005	0.002	0.000	0.002	0.000
Office	0.276	0.265	0.002	0.080	0.001	0.040	0.000
Electrical[a]	0.656	–	–	0.198	0.000	0.178	0.000
Communication[a]	0.264	–	–	0.009	0.000	0.019	0.001
Medical	0.195	0.330	0.000	0.041	0.000	0.034	0.000
Auto	1.370	0.131	0.026	0.207	0.055	0.019	0.004
Other Transport	0.189	0.185	0.000	0.052	0.000	0.005	0.000
Other Manuf	1.522	0.069	0.020	0.012	0.002	0.006	0.001
Aggregate			0.098		0.067		0.009

Notes: $\lvert\Delta \ln \hat{\pi}_{nn}\rvert$ from CP's NAFTA-only general equilibrium counterfactual. Bound: welfare-weighted EK bound $(\widehat{\Delta\varepsilon}_j/(\sigma-1)) \cdot \lvert\Delta \ln \hat{\pi}^j_{nn}\rvert \cdot \alpha^j_n/\theta^j \times 100$ (pp). $\sigma = 4$. [a]Zero baseline domestic share in Mexico; $\Delta \ln \hat{\pi}_{nn}$ undefined.

C Gravity Diagnostic

We apply the diagnostic of Section 4 to the CEPII Gravity database (Conte et al., 2022) at both the aggregate and sectoral level, using bilateral trade flows from BACI (Gaulier and Zignago, 2010), geodesic distances, and GDP for the 30 largest economies by average GDP over 2015–2019, yielding 870 bilateral pairs per year (30×29). We construct domestic expenditure shares as $\lambda_{it} = 1 - \sum_{j \neq i} X_{ijt}/\text{GDP}_{it}$; the mean λ is approximately 0.78, and the median year-to-year change is $|\Delta \ln \lambda| = 0.008$. For each year, we estimate Equations (13) and (14) by PPML with exporter and importer fixed effects, controlling for contiguity, common official language, colonial relationship, and regional trade agreements, and clustering standard errors at the pair level. The sectoral diagnostic uses bilateral trade from BACI HS02 for 2019, estimated separately for each of 96 HS2 sectors with at least 100 bilateral pairs.

At the aggregate level, the curvature coefficient $\hat{\beta}_2$ is statistically insignificant in all years ($p > 0.23$), consistent with log-linear gravity. The implied bounds are small: 0.04–0.08 pp (EK) and 0.06–0.14 pp (Melitz). At the sectoral level, 38 out of 96 sectors (40%) reject $H_0\colon \beta_2 = 0$ at 5%. Departures are concentrated in agriculture and primary products; differentiated manufactures exhibit no significant departures. These finer-grained HS2 results complement the ISIC-level diagnostic used in the CP application (Section 6).

HS2-level diagnostic. Table C1 reports aggregate bounds across $\sigma \in \{3, 4, 5\}$. Table C2 reports the aggregate diagnostic without bilateral controls. Table C3 reports the full sectoral results sorted by p-value. Table C4 reports the sectoral diagnostic without bilateral controls.

Table C1: Aggregate Bounds by σ

	EK bound (pp)			Melitz bound (pp)		
Year	$\sigma = 3$	$\sigma = 4$	$\sigma = 5$	$\sigma = 3$	$\sigma = 4$	$\sigma = 5$
2015	0.11	0.07	0.06	0.18	0.12	0.09
2016	0.10	0.07	0.05	0.17	0.11	0.09
2017	0.07	0.04	0.03	0.11	0.08	0.06
2018	0.06	0.04	0.03	0.09	0.06	0.05
2019	0.12	0.08	0.06	0.20	0.14	0.10

Notes: EK bound from Equation (17); Melitz bound from Equation (18). Both evaluated at $|\Delta \ln \hat{\lambda}| = 0.008$. Both bounds scale as $1/(\sigma - 1)$.

Table C2 reports the aggregate diagnostic without bilateral controls, using only log

distance as in the baseline gravity specification. The curvature coefficient $\hat{\beta}_2$ remains insignificant in all years, confirming that the main results are not driven by the inclusion of bilateral controls.

Table C2: Aggregate Gravity Diagnostic without Bilateral Controls

			Elasticity range		ACR bound (pp)	
Year	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$ (p-value)	$\varepsilon_{\min}$	$\varepsilon_{\max}$	EK	Melitz
2015	0.788	−0.022 (0.52)	0.668	0.879	0.06	0.07
2016	0.778	−0.025 (0.47)	0.645	0.879	0.06	0.08
2017	0.784	−0.015 (0.67)	0.704	0.844	0.04	0.05
2018	0.782	−0.011 (0.76)	0.723	0.826	0.03	0.04
2019	0.769	−0.022 (0.53)	0.653	0.858	0.06	0.07

Notes: PPML with exporter and importer fixed effects, without bilateral controls; pair-clustered standard errors. $\hat{\beta}_2$: coefficient on $(\ln d_{ij})^2$; p-values test H_0: $\beta_2 = 0$. Bounds evaluated at $\sigma = 4$ and $|\Delta \ln \hat{\lambda}| = 0.008$. Sample: 30 countries, 870 pairs/year.

Table C3 reports the gravity diagnostic for all HS2 sectors with at least 100 bilateral pairs, sorted by p-value. Sectors with $p < 0.05$ are indicated by †.

Table C3: Full Sectoral Gravity Diagnostic (HS2, 2019)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
12†	Oil seeds	776	0.417	0.480	<0.001	1.24	2.96
67†	Feathers/down	692	0.455	0.385	<0.001	0.99	2.19
42†	Leather articles	796	0.166	0.204	<0.001	0.53	3.17
03†	Fish	704	0.501	0.269	<0.001	0.69	1.38
07†	Edible vegetables	744	1.152	−0.233	<0.001	0.60	0.52
95†	Toys/games	792	0.498	0.188	<0.001	0.49	0.97
93†	Arms/ammunition	662	−0.463	0.408	<0.001	1.05	-2.27^b
80†	Tin	625	0.612	−0.297	<0.001	0.77	1.25
52†	Cotton	742	0.384	0.308	<0.001	0.79	2.07
15†	Fats/oils	790	0.818	0.294	<0.001	0.76	0.93
72†	Iron/steel	784	0.955	−0.142	<0.001	0.37	0.38
09†	Coffee/tea/spices	787	0.701	0.178	<0.001	0.46	0.66
33†	Perfumery	799	0.743	−0.159	<0.001	0.41	0.55
06†	Live plants	652	1.145	−0.200	<0.001	0.52	0.45

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Table C3: Full Sectoral Gravity Diagnostic (continued)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
75†	Nickel	726	0.248	−0.263	<0.001	0.68	2.74
56†	Wadding/felt	759	0.830	−0.127	<0.001	0.33	0.39
64†	Footwear	793	0.623	0.176	0.001	0.45	0.73
59†	Coated fabrics	758	0.665	−0.104	0.003	0.27	0.40
49†	Printed books	801	0.581	0.160	0.004	0.41	0.71
23†	Food residues	730	0.667	0.139	0.008	0.36	0.54
11†	Milling products	756	0.727	−0.116	0.016	0.30	0.41
05†	Other animal prod.	671	0.364	0.115	0.020	0.30	0.82
34†	Soap/wax	798	0.906	−0.068	0.023	0.18	0.19
27†	Mineral fuels	791	1.036	−0.131	0.023	0.34	0.33
74†	Copper	793	0.968	−0.175	0.024	0.45	0.47
84†	Machinery	808	0.556	−0.062	0.025	0.16	0.29
26†	Ores	699	0.998	−0.119	0.033	0.31	0.31
31†	Fertilizers	703	0.788	0.122	0.034	0.32	0.40
97†	Art/antiques	768	0.260	−0.206	0.036	0.53	2.04
22†	Beverages	793	0.592	−0.082	0.036	0.21	0.36
79†	Zinc	715	1.010	−0.140	0.037	0.36	0.36
35†	Albuminoidal subst.	773	0.704	−0.060	0.039	0.16	0.22
91†	Clocks/watches	777	0.336	0.192	0.040	0.50	1.47
71†	Precious metals	793	0.242	−0.251	0.041	0.65	2.68
02†	Meat	580	0.703	0.157	0.043	0.41	0.58
14†	Vegetable plaiting	585	0.463	−0.186	0.046	0.48	1.04
47†	Paper pulp	643	0.840	−0.151	0.046	0.39	0.46
46†	Straw manufactures	637	0.516	0.143	0.049	0.37	0.72
86	Railway vehicles	741	0.729	−0.166	0.051	0.43	0.59
66	Umbrellas	659	0.769	0.100	0.052	0.26	0.33
62	Woven apparel	801	0.639	0.097	0.052	0.25	0.39
90	Optical/medical	806	0.513	−0.068	0.058	0.18	0.34
32	Tanning/dyeing	788	0.841	−0.054	0.058	0.14	0.17
10	Cereals	694	1.301	0.180	0.060	0.46	0.36
51	Wool	672	0.603	−0.197	0.066	0.51	0.84
19	Cereal preparations	788	0.803	−0.097	0.068	0.25	0.31
08	Edible fruit	752	1.164	−0.115	0.069	0.30	0.26
01	Live animals	667	1.003	−0.149	0.070	0.38	0.38

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Table C3: Full Sectoral Gravity Diagnostic (continued)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
54	Man-made filaments	756	0.663	−0.100	0.074	0.26	0.39
92	Musical instruments	750	0.354	0.094	0.075	0.24	0.69
16	Meat/fish prep.	714	0.919	0.088	0.086	0.23	0.25
29	Organic chemicals	796	0.643	−0.060	0.089	0.16	0.24
65	Headgear	783	0.504	0.070	0.092	0.18	0.36
39	Plastics	807	0.791	−0.052	0.103	0.13	0.17
13	Lac/gums/resins	714	0.386	0.077	0.106	0.20	0.51
58	Special woven fabrics	745	0.800	−0.098	0.121	0.25	0.32
18	Cocoa	765	0.746	−0.085	0.134	0.22	0.29
24	Tobacco	677	0.752	0.161	0.159	0.41	0.55
04	Dairy	713	1.014	−0.064	0.169	0.17	0.16
36	Explosives	604	0.354	0.080	0.194	0.21	0.59
81	Other base metals	729	0.436	−0.057	0.217	0.15	0.34
44	Wood	801	0.504	0.073	0.219	0.19	0.37
38	Chemical products	793	0.720	−0.031	0.222	0.08	0.11
68	Stone/cement art.	785	0.849	0.040	0.292	0.10	0.12
28	Inorganic chemicals	787	0.802	−0.044	0.313	0.11	0.14
78	Lead	638	0.893	−0.085	0.313	0.22	0.25
82	Tools	806	0.565	0.035	0.329	0.09	0.16
60	Knitted fabrics	707	0.725	−0.094	0.340	0.24	0.34
48	Paper	801	0.934	−0.031	0.344	0.08	0.09
63	Other textiles	802	0.867	0.047	0.381	0.12	0.14
17	Sugars	773	0.875	−0.040	0.508	0.10	0.12
61	Knitted apparel	801	0.834	0.038	0.512	0.10	0.12
45	Cork	625	1.010	−0.045	0.563	0.12	0.11
89	Ships	705	0.591	−0.049	0.644	0.13	0.21
70	Glass	795	0.819	−0.018	0.644	0.05	0.06
55	Man-made staple	741	0.487	0.027	0.651	0.07	0.14
21	Misc food prep.	799	0.690	0.016	0.678	0.04	0.06
69	Ceramics	782	0.672	0.013	0.712	0.03	0.05
85	Electrical machinery	810	0.686	0.013	0.720	0.03	0.05
94	Furniture	794	0.715	0.013	0.749	0.03	0.05
83	Misc base metal art.	792	0.710	−0.013	0.754	0.03	0.05
53	Vegetable fibers	640	0.562	−0.033	0.755	0.09	0.15

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Table C3: Full Sectoral Gravity Diagnostic (continued)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
76	Aluminum	801	0.749	−0.010	0.797	0.03	0.03
40	Rubber	804	0.607	0.009	0.811	0.02	0.04
50	Silk	567	0.414	0.030	0.819	0.08	0.19
30	Pharmaceuticals	795	0.316	0.012	0.820	0.03	0.10
43	Furskins	664	0.158	0.023	0.839	0.06	0.37
87	Vehicles	798	0.624	0.009	0.841	0.02	0.04
25	Salt/stone/cement	788	1.076	−0.008	0.846	0.02	0.02
96	Misc manufactured	793	0.612	0.004	0.909	0.01	0.02
41	Hides/skins	735	0.746	0.007	0.911	0.02	0.02
57	Carpets	745	0.979	0.005	0.918	0.01	0.01
20	Vegetable prep.	778	0.806	−0.003	0.958	0.01	0.01
88	Aircraft	771	0.148	−0.002	0.977	<0.01	0.03
37	Photographic goods	663	0.592	0.002	0.981	<0.01	0.01
73	Iron/steel articles	804	0.754	−0.000	0.998	<0.01	<0.01

†: $p < 0.05$. PPML with exporter and importer fixed effects, controlling for contiguity, common official language, colonial relationship, and regional trade agreements; pair-clustered standard errors. Bounds evaluated at $\sigma = 4$ and $|\Delta \ln \hat{\lambda}| = 0.008$. EK bound from Equation (17); Melitz bound from Equation (18). [b]Negative pooled elasticity; Melitz bound not interpretable.

Table C4 reports the sectoral gravity diagnostic without bilateral controls. Out of 96 sectors, 39 (41%) reject $H_0\colon \beta_2 = 0$ at 5%, compared with 38 (40%) in the main specification with bilateral controls (Table C3). The qualitative pattern is unchanged.

Table C4: Sectoral Gravity Diagnostic without Bilateral Controls (HS2, 2019)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
07†	Edible vegetables	744	1.453	−0.251	<0.001	0.65	0.44
03†	Fish	704	0.726	0.301	<0.001	0.78	1.07
12†	Oil seeds	776	0.866	0.440	<0.001	1.13	1.31
93†	Arms/ammunition	662	0.320	0.504	<0.001	1.30	4.07
42†	Leather articles	796	0.307	0.222	<0.001	0.57	1.87
80†	Tin	625	0.857	−0.289	<0.001	0.75	0.87

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Table C4: Sectoral Gravity Diagnostic without Bilateral Controls (continued)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
95†	Toys/games	792	0.841	0.211	<0.001	0.54	0.65
23†	Food residues	730	0.976	0.215	<0.001	0.55	0.57
15†	Fats/oils	790	1.034	0.232	<0.001	0.60	0.58
67†	Feathers/down	692	0.879	0.449	<0.001	1.16	1.32
11†	Milling products	756	1.319	−0.176	<0.001	0.45	0.34
09†	Coffee/tea/spices	787	0.915	0.174	<0.001	0.45	0.49
26†	Ores	699	1.408	−0.182	<0.001	0.47	0.33
46†	Straw manufactures	637	0.825	0.321	<0.001	0.83	1.00
52†	Cotton	742	0.750	0.226	0.001	0.58	0.78
06†	Live plants	652	1.340	−0.144	0.001	0.37	0.28
79†	Zinc	715	1.284	−0.176	0.002	0.45	0.35
04†	Dairy	713	1.459	−0.114	0.003	0.29	0.20
44†	Wood	801	0.958	0.177	0.003	0.46	0.48
10†	Cereals	694	1.280	0.210	0.003	0.54	0.42
08†	Edible fruit	752	1.537	−0.190	0.003	0.49	0.32
49†	Printed books	801	1.004	0.209	0.003	0.54	0.54
66†	Umbrellas	659	1.172	0.171	0.003	0.44	0.38
33†	Perfumery	799	0.864	−0.136	0.004	0.35	0.41
75†	Nickel	726	0.475	−0.242	0.006	0.62	1.31
05†	Other animal prod.	671	0.693	0.147	0.006	0.38	0.55
16†	Meat/fish prep.	714	1.152	0.157	0.006	0.41	0.35
64†	Footwear	793	0.870	0.161	0.007	0.42	0.48
18†	Cocoa	765	1.142	−0.121	0.010	0.31	0.27
68†	Stone/cement art.	785	1.023	0.101	0.014	0.26	0.26
56†	Wadding/felt	759	1.012	−0.092	0.018	0.24	0.24
13†	Lac/gums/resins	714	0.524	0.145	0.018	0.37	0.71
74†	Copper	793	1.210	−0.141	0.019	0.36	0.30
62†	Woven apparel	801	0.845	0.132	0.021	0.34	0.40
72†	Iron/steel	784	1.160	−0.083	0.023	0.21	0.18
17†	Sugars	773	1.397	−0.123	0.032	0.32	0.23
97†	Art/antiques	768	0.156	−0.212	0.037	0.55	3.50
92†	Musical instruments	750	0.491	0.109	0.040	0.28	0.57
59†	Coated fabrics	758	0.958	−0.084	0.047	0.22	0.23
82	Tools	806	0.710	0.073	0.051	0.19	0.26

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Table C4: Sectoral Gravity Diagnostic without Bilateral Controls (continued)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
53	Vegetable fibers	640	0.419	0.160	0.055	0.41	0.99
51	Wool	672	0.838	−0.199	0.057	0.51	0.61
71	Precious metals	793	0.342	−0.238	0.062	0.61	1.79
31	Fertilizers	703	1.050	0.080	0.071	0.21	0.20
86	Railway vehicles	741	1.190	−0.121	0.078	0.31	0.26
36	Explosives	604	0.572	0.110	0.084	0.28	0.50
34	Soap/wax	798	1.104	−0.050	0.086	0.13	0.12
22	Beverages	793	0.804	−0.081	0.099	0.21	0.26
63	Other textiles	802	1.104	0.099	0.104	0.25	0.23
90	Optical/medical	806	0.547	−0.055	0.107	0.14	0.26
39	Plastics	807	1.011	−0.049	0.128	0.13	0.13
47	Paper pulp	643	0.979	−0.111	0.138	0.29	0.29
01	Live animals	667	1.699	−0.115	0.150	0.30	0.17
58	Special woven fabrics	745	0.937	−0.097	0.152	0.25	0.27
91	Clocks/watches	777	0.321	0.122	0.161	0.31	0.98
57	Carpets	745	1.104	0.066	0.162	0.17	0.15
54	Man-made filaments	756	0.840	−0.073	0.163	0.19	0.22
19	Cereal preparations	788	1.216	−0.060	0.177	0.16	0.13
84	Machinery	808	0.714	−0.040	0.203	0.10	0.14
35	Albuminoidal subst.	773	0.897	−0.037	0.227	0.09	0.11
29	Organic chemicals	796	0.754	−0.043	0.240	0.11	0.15
14	Vegetable plaiting	585	0.769	−0.100	0.250	0.26	0.34
27	Mineral fuels	791	1.269	−0.068	0.262	0.18	0.14
81	Other base metals	729	0.494	−0.046	0.309	0.12	0.24
65	Headgear	783	0.816	0.059	0.342	0.15	0.19
61	Knitted apparel	801	1.053	0.054	0.373	0.14	0.13
28	Inorganic chemicals	787	0.845	−0.036	0.386	0.09	0.11
32	Tanning/dyeing	788	0.951	−0.024	0.395	0.06	0.07
94	Furniture	794	1.082	0.043	0.406	0.11	0.10
24	Tobacco	677	0.899	0.096	0.413	0.25	0.28
78	Lead	638	1.218	−0.065	0.439	0.17	0.14
73	Iron/steel articles	804	0.991	0.031	0.454	0.08	0.08
02	Meat	580	1.235	0.089	0.456	0.23	0.18
60	Knitted fabrics	707	1.136	−0.058	0.507	0.15	0.13

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Table C4: Sectoral Gravity Diagnostic without Bilateral Controls (continued)

						Bound (pp)	
HS2	Sector	N	$\hat{\varepsilon}_{\text{pool}}$	$\hat{\beta}_2$	p	EK	Melitz
76	Aluminum	801	1.020	0.027	0.521	0.07	0.07
21	Misc food prep.	799	1.025	0.025	0.543	0.06	0.06
55	Man-made staple	741	0.676	0.037	0.546	0.09	0.14
25	Salt/stone/cement	788	1.243	0.024	0.553	0.06	0.05
87	Vehicles	798	0.913	−0.029	0.575	0.08	0.08
41	Hides/skins	735	0.918	−0.033	0.584	0.09	0.09
96	Misc manufactured	793	0.832	0.024	0.628	0.06	0.07
69	Ceramics	782	0.887	0.020	0.640	0.05	0.06
88	Aircraft	771	0.373	0.031	0.655	0.08	0.22
38	Chemical products	793	0.849	−0.011	0.659	0.03	0.03
45	Cork	625	1.301	0.039	0.670	0.10	0.08
48	Paper	801	1.243	−0.012	0.730	0.03	0.02
30	Pharmaceuticals	795	0.350	0.017	0.751	0.04	0.12
50	Silk	567	0.720	0.039	0.771	0.10	0.14
20	Vegetable prep.	778	0.995	−0.012	0.781	0.03	0.03
43	Furskins	664	0.501	−0.037	0.794	0.10	0.19
89	Ships	705	0.843	−0.023	0.818	0.06	0.07
37	Photographic goods	663	0.638	0.013	0.867	0.03	0.05
40	Rubber	804	0.857	−0.005	0.913	0.01	0.01
83	Misc base metal art.	792	0.980	−0.004	0.925	0.01	0.01
85	Electrical machinery	810	0.826	−0.001	0.979	<0.01	<0.01
70	Glass	795	0.994	−0.000	0.998	<0.01	<0.01

†: $p < 0.05$. PPML with exporter and importer fixed effects, distance only, no bilateral controls; pair-clustered standard errors. Bounds evaluated at $\sigma = 4$ and $|\Delta \ln \hat{\lambda}| = 0.008$. EK bound from Equation (17); Melitz bound from Equation (18).